

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester VI Examination, 2022-23

**Paper: GLB-603 — Hydrogeology, Environmental Geology,
Exploration Geology and Computer Application**

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 058442

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. All questions are of equal value.*

SECTION — A

Question 1. Describe Darcy's law. A confined aquifer is 30 m thick and 7 km wide. Two observation wells are located 1.0 km apart in the direction of flow. The head in well *A* is 97.0 m and in well *B* it is 92.0 m. If the total daily flow of water through the aquifer is 1050 m³/day, then calculate the hydraulic conductivity of the aquifer. [14]

Question 2. Describe the basic terms related to aquifer properties. Calculate groundwater storage for an aquifer having an areal extent of 10 km² and saturated thickness 15 m. Consider the specific yield of the aquifer is 20%. [14]

Question 3. Explain the processes and theories used to explain monsoons. Detail their significance in the Indian context as well. [14]

Question 4. What are landslides? Explain in detail with their mechanism and classification. What are the global air circulation patterns and how do they affect the climate? [14]

SECTION — B

Question 5. Write short notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Ghyben-Herzberg equation of seawater intrusion
- (b) Origin of different types of springs
- (c) Hydrochemical properties used to characterize groundwater quality
- (d) Thornthwaite classification
- (e) Runoff process

Question 6. What are the different exploration methods used for mineral exploration? How are electrical resistivity methods used for exploration? Define the difference between profiling and sounding. [14]

Question 7. Explain the fundamental principle of electromagnetic induction with a suitable diagram. What is the self-potential method? Discuss how you acquire SP data and its interpretation. [14]

SECTION — C

Question 8. Define or explain the following in brief:

[7 × 2 = 14]

- (a) Effluent and influent conditions
- (b) Water level and water table
- (c) Constant rate of infiltration
- (d) Electrical logging
- (e) Assay techniques
- (f) Global warming
- (g) Intensity of earthquakes